

Important Git commands	Description	Command
	Set a user name	<code>git config --global user.name "Mitesh"</code>
	Set email address	<code>git config --global user.email "xxxxxxxx@xxxxx.com"</code>
	Initialise a Git repository	<code>git init</code>
	Clone a repository	<code>git clone https://github.com/xxxxxxx/Angular9786-DevOps</code>
	Verify the status of files in the repository	<code>git status</code>
	Add files in the current directory for commit operation	<code>git add</code>
	Commit files	<code>git commit -m "First commit in the repository"</code>
	List your branches	<code>git branch</code>
	Switch branch	<code>git checkout</code>
	Create a new branch	<code>git branch new-branch-name</code>
	Merge current branch with the target branch	<code>git merge target-branch</code>
	Show all commits	<code>git log</code>
	Save modified and staged changes	<code>git stash</code>
	Fetch and merge any commits from the remote branch	<code>git pull</code>

2. Jenkins

Jenkins is an open source tool that provides integration with the existing tools used in application life cycle management to automate all activities -- build, test, and deploy. It is almost a synonym for DevOps, and is one of the easiest to learn. It has a huge community for support. We can easily install Jenkins in the Kubernetes cluster and utilise multiple cloud services to make it highly available.

Jenkins provides a simple way to create a pipeline that includes continuous integration and continuous delivery. It can be used creatively for managing entire life cycle management activities easily. Plugin based architecture helps us to use the existing tools in the pipeline and hence too many changes in culture can be avoided in one go.

Initial release	2011
Stable release	2.375.1 LTS. More information at https://www.jenkins.io/changelog-stable/
Written in	Java
Licence	MIT
Website	www.jenkins.io
GitHub repository	https://github.com/jenkinsci/jenkins
Features	- Jenkins installation is supported for Docker, FreeBSD, Gentoo, MacOS X, OpenBSD, OpenSUSE, Red Hat/Fedora/CentOS, Ubuntu/Debian, Windows, and Generic Java package (.war) - Multiple views are supported along with folders to manage jobs in an easier manner - New UI for Jenkins dashboard with ease of use - New user interface to create Pipeline as a Code in an easier way - Pipeline as a Code/multi-branch pipeline to manage pipelines easily in repositories - Distributed architecture to manage Jenkins effectively - Authorisation: Role-based access to users based on build jobs/folders/pipelines/configurations/actions - Authentication: Integration with active directory, LDAP and with internal database
How is it useful in DevOps practices implementation?	Automation server that helps to automate activities involved in application life cycle management
Can we integrate it with Pipeline as Code?	Pipeline as Code using Jenkinsfile - Declarative pipeline - Scripted pipeline - Pipeline as YAML - Blue Ocean
Is a commercial flavour available?	CloudBees Jenkins
Awards	- DevOps Dozen 2017: Best DevOps Open Source Project - DevOps Dozen 2016: Best DevOps Open Source Project - Bossie Awards 2014: The best open source application development tools - Geek Choice Award 2014 - SDTimes 100, 2014: DevOps & SCM - SDTimes 100, 2013: Development Tools & CI - SDTimes 100, 2012: Development Tools & CI - InfoWorld Bossies Award, 2011 - O'Reilly Open-Source Award, 2011 - SDTimes 100, 2011: ALM&SCM - SDTimes 100, 2010: ALM Platforms - GlassFish Community Innovation Award 2008 - Duke's Choice Award